

# Simulation Visualization Dashboard

## Overview

Panisim provides a web-based visualization dashboard to monitor vessel simulation in real-time. The dashboard offers a comprehensive interface for monitoring vessel state, visualizing trajectories, viewing sensor data, and interacting with the simulation.

The visualization system consists of two main components:

1. **2D Dashboard** - A data-rich interface showing vessel state, charts, and control inputs
2. **3D Visualization** - A three-dimensional view of the vessel in its environment with position and orientation

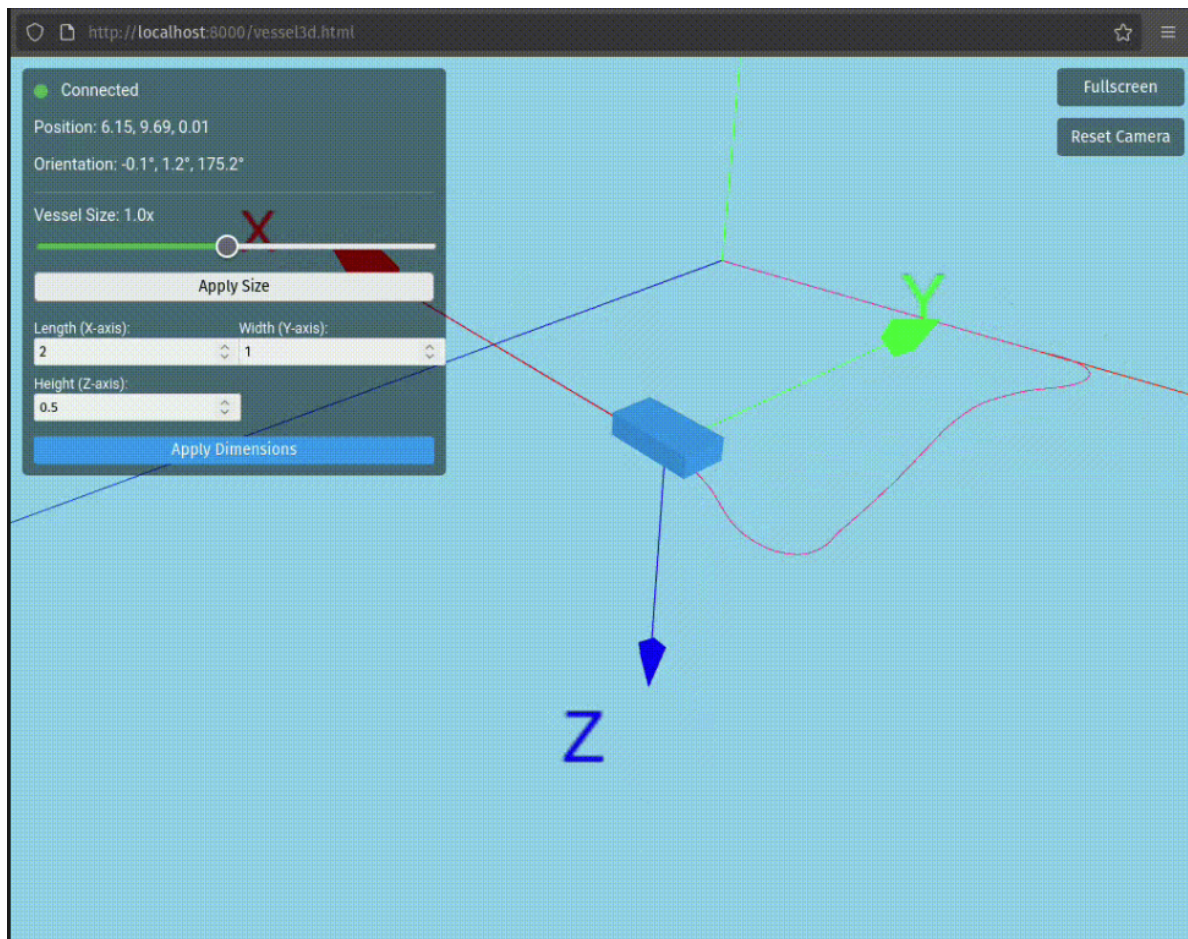
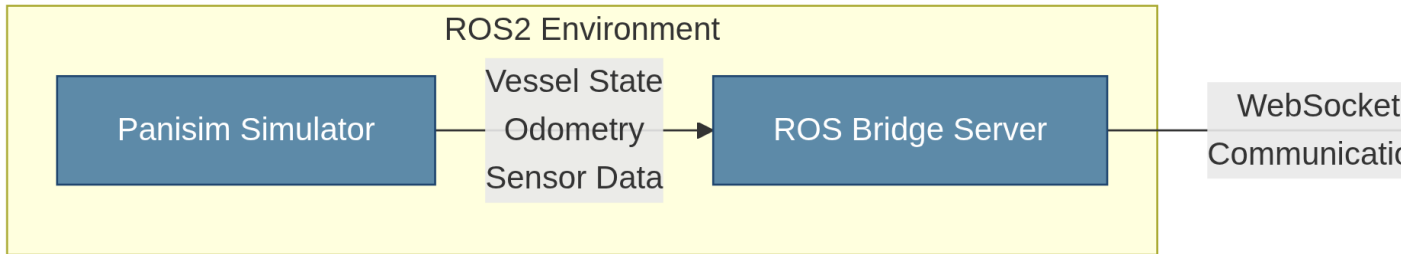


Figure 1: Way-point trackingSimulation Visualization



## Starting the Visualization

The web interface is automatically started when launching the Panisim simulator. It uses:

1. A lightweight HTTP server to host the web files
2. The ROS Bridge server for WebSocket communication with ROS2

The visualization launches when running the simulation:

```
ros2 launch gnc gnc_launch.py
```

Once started, you can access the dashboard by opening your web browser:

- **Main Dashboard:** <http://localhost:8000/>
- **3D Visualization:** <http://localhost:8000/vessel3d.html>

## Main Dashboard

The main dashboard (index.html) provides a comprehensive overview of the vessel's state, sensor data, and control inputs.

## Key Features

1. **Vessel Selection:** Switch between different vessels in the simulation
2. **Connection Status:** Real-time display of connectivity to the ROS2 environment
3. **Vessel Information:** Display of key vessel parameters (position, velocity, orientation)
4. **Data Visualizations:** Real-time charts for:
  - Vessel path and trajectory
  - Velocities (linear and angular)
  - Orientation (roll, pitch, yaw)
  - Control surface positions
  - Thruster states

## Dashboard Layout

The dashboard is organized in a responsive grid layout with cards for different data categories:

| Section                     | Description                                                             |
|-----------------------------|-------------------------------------------------------------------------|
| <b>Vessel Info</b>          | Basic vessel parameters including position, velocities, and orientation |
| <b>Path Visualization</b>   | 2D trajectory visualization showing vessel movement in X-Y plane        |
| <b>Vessel Velocities</b>    | Charts for surge (u), sway (v), and heave (w) velocities                |
| <b>Vessel Angular Rates</b> | Charts for roll rate (p), pitch rate (q), and yaw rate (r)              |
| <b>Orientation</b>          | Charts for roll ( ), pitch ( ), and yaw ( ) angles                      |
| <b>Control Surfaces</b>     | Visual representation of control surface positions                      |
| <b>Thrusters</b>            | Visual representation of thruster RPMs                                  |

## Real-time Data Updates

The dashboard updates in real-time with configurable update rates. Data collection begins automatically when connecting to a vessel and includes:

- Subscribe to `/[vessel_name]/odometry_sim` for position and velocity data
- Subscribe to `/[vessel_name]/vessel_state` for complete vessel state including actuators
- Optional subscription to additional sensor topics

## 3D Visualization

The 3D visualization (vessel3d.html) provides an interactive three-dimensional view of the vessel in its environment.

### Key Features

1. **Real-time Movement:** The vessel model moves and rotates according to simulation data
2. **Trajectory Path:** A colored trail showing the vessel's path through the environment
3. **Customizable View:** Camera controls for rotating, panning, and zooming
4. **Vessel Dimensions:** Adjustable vessel size and proportions to match different vessel types
5. **Full-screen Mode:** Expand the visualization to utilize the entire screen

### Controls

The 3D view offers several interactive controls:

- **Orbit:** Click and drag to rotate the camera around the vessel
- **Pan:** Right-click and drag (or middle-click and drag) to move the camera
- **Zoom:** Scroll wheel to zoom in and out
- **Reset Camera:** Button to restore the default camera position
- **Fullscreen:** Button to toggle fullscreen mode

### Vessel Model Customization

You can adjust the vessel's appearance through the control panel:

- **Vessel Size:** Overall scaling of the vessel model
- **Dimensions:** Individual control over length, width, and height proportions
- **Apply Changes:** Button to update the model with the new dimensions

## Integration with ROS2

The visualization dashboard communicates with ROS2 using the ROS Bridge Server, which provides WebSocket connectivity between the ROS2 environment and web clients.

## Communication Flow

1. **ROS2 Topics to WebSockets:** The ROS Bridge Server converts ROS2 topic messages to WebSocket messages
2. **Web Client Processing:** JavaScript code processes the WebSocket messages and updates the visualizations
3. **Bidirectional Communication:** The system supports both subscribing to and publishing ROS2 topics

## Required ROS2 Components

To enable the visualization, the launch file includes:

```
# HTTP server for web files
ExecuteProcess(
  cmd=['python3', '-m', 'http.server', '8000', '--directory', web_build_dir],
  name='http_server',
  output='screen'
),

# ROS Bridge server for WebSocket communication
IncludeLaunchDescription(
  AnyLaunchDescriptionSource(rosbridge_launch)
),
```

## Topic Subscriptions

The visualization automatically discovers vessel topics by querying available ROS2 topics. It looks for the following patterns:

- `/<vessel_name>/odometry_sim`: For vessel position and velocity
- `/<vessel_name>/vessel_state`: For complete vessel state including actuators

## Implementation Details

The visualization dashboard is built using modern web technologies:

- **Chart.js**: For responsive, interactive data visualization
- **three.js**: For 3D rendering of the vessel
- **roslibjs**: For ROS2 communication via WebSockets
- **Responsive Design**: Automatically adapts to different screen sizes

## Key JavaScript Components

The system is organized into the following components:

1. **VesselManager**: Handles vessel discovery, selection, and configuration
2. **VesselVisualizer**: Manages data collection, processing, and visualization
3. **3D Rendering**: Handles the three-dimensional visualization of vessels

## Vessel State Representation

The dashboard visualizes the complete vessel state model:

- **Position**:  $x, y, z$  (in meters)
- **Linear Velocity**:  $u, v, w$  (in m/s)
- **Angular Velocity**:  $p, q, r$  (in rad/s)
- **Orientation**:  $\phi, \theta, \psi$  (in degrees)
- **Actuators**: Control surface angles (in degrees) and thruster RPMs

## Customization Options

The visualization can be customized in several ways:

1. **Update Rates**: Chart refresh rates can be adjusted
2. **Data Window**: The amount of historical data displayed in charts
3. **Vessel Configuration**: The number of control surfaces and thrusters displayed
4. **3D Model**: Vessel dimensions and appearance
5. **Chart Scaling**: Auto-rescaling and manual zoom/pan on charts

## Troubleshooting

### Connection Issues

If the dashboard shows “Disconnected” status:

1. Ensure the ROS Bridge server is running
2. Check that WebSocket port 9090 is accessible
3. Verify the simulator is publishing required topics

## Missing Vessel Data

If vessel data isn't displaying:

1. Select a vessel from the dropdown menu
2. Check ROS2 topics are being published:

```
ros2 topic list | grep vessel
```

3. Verify message publication with:

```
ros2 topic echo /<vessel_name>/odometry_sim
```

## Browser Compatibility

The visualization works best with modern browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Brave Browser

## Conclusion

The Panisim visualization dashboard provides a comprehensive interface for monitoring and interacting with simulated vessels. With its real-time updates, interactive charts, and 3D visualization, it offers a powerful tool for understanding vessel behavior and validating custom control or guidance algorithms.